The Holographic Count Equation and the Decomposition of $4\pi^2$

A Native SFT Action Principle and the Two-Factor Structure of the Friedmann–SFT Geometric Ratio

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Abstract

We present two results that strengthen the mathematical foundations of Solvency Field Theory. First, we derive the SFT modified Friedmann equation $H^2 = 2G\rho/(3\pi)$ from a holographic count equation—the postulate that the number of Planck-area cells on the Hubble horizon equals the number of complete Compton phase cycles performed by all bulk matter during one Hubble maintenance interval. This derivation requires no free parameters and no unexplained proportionality constants. We formulate the corresponding constrained minisuperspace action principle. Second, we provide a rigorous decomposition of the $4\pi^2$ ratio between SFT and standard Friedmann into two independently identifiable factors of 2π: one from the linear-versus-angular phase convention (h vs $\hbar$), and one from the boundary-versus-bulk normalization (Bekenstein-Hawking horizon accounting vs Einstein field equation coupling). This decomposition replaces earlier, less precise attributions and establishes the $4\pi^2$ as a product of two specific unit-conversion layers, not a single unexplained geometric factor.

1. Motivation

In V7.1, the Compton maintenance postulate was shown to produce the modified Friedmann equation $H^2 = 2G\rho/(3\pi)$ through algebraic substitution into the master equation $H = \dot{S}/I$. That derivation is correct but does not constitute a variational principle. It does not specify the mathematical object that the theory extremizes, nor does it explain the origin of the $4\pi^2$ ratio beyond attributing it to "the geometry of mapping flat-space ticking onto a spherical boundary."

This note addresses both gaps. Section 2 introduces the holographic count equation and derives the modified Friedmann equation from it. Section 3 formulates the constrained action principle. Section 4 provides the rigorous decomposition of $4\pi^2$ into two independent 2π factors. Section 5 establishes the Planck scale as the unique minimax solvent resolution. Section 6 identifies the remaining open problems.

2. The Holographic Count Equation

2.1 Definitions

Horizon cell inventory. The number of Planck-area cells on the Hubble horizon:

$$\mathbf{N_h = A_h / (4\ell_p^2)} \quad (1)$$

where $A_H = 4\pi(c/H)^2$ is the Hubble horizon area and $\ell_P^2 = \hbar G/c^3$ is the Planck area.

Bulk maintenance count. The total number of complete Compton phase cycles performed by all bulk matter during one Hubble maintenance interval $t_H = 1/H$:

$$\mathbf{N_{ti}c^k = E_h / (hH)} \quad (2)$$

where $E_H = \rho c^2 V_H$ is the total rest energy in the Hubble volume, h is Planck's constant (linear, not angular), and 1/H is one Hubble time—the maximum causal communication timescale.

2.2 The postulate

The Holographic Count Equation. *The number of boundary cells on the Hubble horizon equals the number of complete bulk maintenance cycles accumulated during one causal maintenance time:*

$$\mathbf{A_h / (4\ell_p^2) = E_h / (hH)} \quad (3)$$

The ledger balance condition: every bit of bulk maintenance is accounted for by one cell of boundary capacity, and every cell of boundary capacity corresponds to one bit of bulk maintenance, settled over one Hubble interval.

The choice of the Hubble time as the maintenance interval is physically motivated: it is the timescale over which a signal can traverse the Hubble radius c/H. In flat FLRW spacetime, the Hubble sphere coincides with the apparent horizon—the horizon that appears in rigorous cosmological thermodynamic derivations. The count equation is therefore an apparent-horizon thermodynamic balance.

2.3 Derivation in flat FLRW

For a flat FLRW spacetime with Hubble parameter H, homogeneous density ρ:

Horizon area: $A_H = 4\pi(c/H)^2$. Hubble volume: $V_H = (4\pi/3)(c/H)^3$. Total energy: $E_H = \rho c^2 V_H = (4\pi/3)\rho c^5/H^3$.

Left-hand side (horizon cell count), using $\ell_P^2 = \hbar G/c^3$:

$$\mathbf{N_h = \pi c^5 / (\hbar G H^2)} \quad (4)$$

Right-hand side (bulk maintenance count), using $h = 2\pi\hbar$:

$$\mathbf{N_{ti}c^k = 2\rho c^5 / (3\hbar H^4)} \quad (5)$$

Equating $N_H = N_{tick}$ and cancelling common factors (c^5, ħ):

$$\mathbf{\pi / (GH^2) = 2\rho / (3H^4)} \quad (6)$$

Solving for H^2:

$$\mathbf{H^2 = 2G\rho / (3\pi)} \quad (7)$$

This is exactly the V7.1 modified Friedmann equation, derived with **zero free parameters**. No proportionality constant γ was introduced or required.

2.4 Why the instantaneous rate law required γ = 3

A previous attempt formulated the SFT law as an instantaneous rate equation: $d/dt(A_H/4\ell_P^2) = \gamma \times E_H/h$. This produced $H^2 = 2\gamma G\rho/(9\pi) + C$, requiring $\gamma = 3$ by hand. The count equation resolves this by recognizing that the theory has **memory over one causal interval**. The ledger is settled over one Hubble time, not instantaneously. The factor of 1/H in the count equation carries exactly the volume-to-area information that eliminates the need for γ.

3. The Constrained Action Principle

The holographic count equation can be promoted to a variational principle by introducing a Lagrange multiplier Λ(t) that enforces the ledger constraint:

$$\mathbf{S_eff = \int dt\, \Lambda(t)\, [A_h(a,H) / (4\ell_p^2) - E_h(a,\rho) / (hH)] + S_{matter}} \quad (8)$$

where a(t) is the FLRW scale factor, $H = \dot{a}/a$, and S_{matter} ensures the continuity equation $\dot{\rho} + 3H\rho = 0$ for pressureless dust.

This is a **constrained minisuperspace action**. It is not a standard local Lagrangian density integrated over spacetime. It is a holographic constraint enforced at the boundary level. The fundamental law relates bulk content to boundary geometry—consistent with the information-theoretic character of the framework.

The Lagrange multiplier Λ(t) is the "ledger price"—the cost of violating the count balance. At the classical level, Λ is determined by the constraint. The physical content lives in the constraint equation, not in the multiplier. This is analogous to how the lapse function in canonical GR enforces the Hamiltonian constraint without contributing independent dynamics.

4. The Rigorous Decomposition of $4\pi^2$

The ratio between the standard Friedmann equation ($H^2 = 8\pi G\rho/3$) and the SFT count equation ($H^2 = 2G\rho/(3\pi)$) is exactly $4\pi^2 = (2\pi)^2 \approx 39.48$. In V7.1, this was attributed broadly to "the distinction between angular and linear units of quantum phase projected onto a spherical boundary." That attribution was directionally correct but insufficiently precise. This section provides the rigorous decomposition.

4.1 Factor 1: Phase convention (h vs ħ)

The count equation uses the linear Planck constant h in the bulk tick rate (Eq. 2), while the horizon capacity uses the angular Planck constant ħ through the Planck area $\ell_P^2 = \hbar G/c^3$ (Eq. 1). Since $h = 2\pi\hbar$, this introduces one factor of 2π.

To isolate this factor, consider replacing h with ħ in the count equation—counting one tick per radian of phase advance rather than one tick per complete cycle. The modified bulk count becomes $N'_{tick} = E_H/(\hbar H)$, and the resulting Friedmann-type equation is:

$$\mathbf{H^2 = 4G\rho / 3} \quad (9)$$

Compared to standard Friedmann ($H^2 = 8\pi G\rho/3$), this differs by a factor of 2π, not $4\pi^2$. The first factor of 2π is therefore cleanly attributable to the choice of complete phase cycles (h) versus angular phase (ħ) in the definition of one irreversible bit.

This factor is empirically confirmed: the SPARC rotation curve test (V7.1) shows that $cH_0/(2\pi)$ outperforms cH_0 across 137 galaxies. The data select the complete-cycle convention.

4.2 Factor 2: Boundary vs bulk normalization

Even with the angular convention (ħ), the count equation gives $H^2 = 4G\rho/3$ while Friedmann gives $H^2 = 8\pi G\rho/3$. The remaining factor of 2π is independent of the phase convention. It arises from a different source: the normalization conventions used by SFT and GR to relate geometry to content.

On the SFT side, the horizon capacity is:

$$\mathbf{N_h = A / (4\ell_p^2)} \quad (10)$$

The factor of 4 in the denominator comes from the Bekenstein-Hawking entropy formula $S = A/(4\ell_P^2)$. This is a thermodynamic normalization: it sets the number of independent degrees of freedom on the horizon.

On the GR side, the Einstein field equation carries a coupling coefficient 8πG:

$$\mathbf{G\mu_\nu = 8\pi G\, T\mu_\nu} \quad (11)$$

The 8π is not derived from first principles within GR—it is fixed by requiring consistency with the Newtonian weak-field limit and the Poisson equation $\nabla^2\Phi = 4\pi G\rho$.

The ratio between these two normalization conventions produces the second 2π. When the SFT count equation (using $A/4\ell_P^2$) is compared to the Friedmann equation (using 8πG), the ratio of their geometric coefficients is:

$$\mathbf{[2G\rho/(3\pi)] \div [4G\rho/3] = 1/(2\pi) \text{ per phase factor}} \quad (12)$$

or equivalently, the full Friedmann coefficient 8πG/3 contains a factor of 2π relative to the count equation's coefficient 4G/3 that is not attributable to the phase convention. This factor reflects the different way SFT (boundary thermodynamics) and GR (bulk field equations) normalize the coupling between geometry and matter.

4.3 The complete factorization

The full $4\pi^2$ decomposes cleanly:

$$\mathbf{4\pi^2 = 2\pi \text{ (phase convention)} \times 2\pi \text{ (boundary/bulk normalization)}} \quad (13)$$

Factor	Origin	SFT uses	GR uses
First 2π	Phase convention: what counts as one bit	h (complete cycles)	ħ (radians)
Second 2π	Geometry coupling: how content maps to geometry	$A/(4\ell_p^2)$ boundary	8πG bulk coupling
Product	Total ratio: H^2_Friedmann / H^2_SFT	$\mathbf{4\pi^2 \approx 39.48}$	*(content-independent)*

Table 1: The two-factor decomposition of $4\pi^2$. Each factor has a distinct and independently identifiable origin. Neither depends on the matter content of the universe.

4.4 What this decomposition replaces

This decomposition supersedes two earlier attributions:

V7.1 attribution: "One factor of 2π from the frequency conversion, and a second from projecting flat-space results onto a spherical boundary." This was directionally correct for the first factor but vague for the second. The second factor does not come from spherical projection; it comes from the different normalization conventions of boundary thermodynamics and bulk field equations.

Local-vs-global hypothesis: An earlier conjecture attributed the $4\pi^2$ to the ratio between locally-sourced expansion (within one Hubble volume) and globally-sourced expansion (from all of homogeneous space). This was shown to be inconsistent with Birkhoff's theorem: in homogeneous FLRW, the Friedmann equation is already local. The standard Friedmann coefficient does not receive additional contributions from matter beyond the Hubble sphere. The $4\pi^2$ is a normalization ratio, not a source ratio.

5. The Planck Scale as Minimax Solvent Resolution

The Planck area ℓ_P^2 enters the count equation as the boundary cell size. Its appearance is not assumed—it is the unique scale at which quantum localization and gravitational confinement are simultaneously satisfiable. For an excitation of mass m:

Quantum bound: $L_Q \sim \hbar/(mc)$ Gravitational bound: $L_G \sim Gm/c^2$

A configuration is solvent—simultaneously quantum-localizable and gravitationally stable—only if $L \geq \max(L_Q, L_G)$. Minimizing the required size over all m yields:

$$\mathbf{m^* = m_p = \sqrt{(\hbar c/G)}, \quad L^* = \ell_p = \sqrt{(\hbar G/c^3)}} \quad (14)$$

The Planck scale is the **minimax solvent resolution**: the smallest cell size at which both quantum and gravitational consistency are simultaneously maintained. This provides a first-principles justification for the boundary tiling unit in Eq. (1).

6. Comparison of Formulations

Formulation	Equation	Free parameters	Produces $H^2 = 2G\rho/(3\pi)$?
V7.1 master equation	$H = \dot{S}/I$	None	Yes (algebraic)
Instantaneous rate law	$d/dt(N_h) = \gamma \times \dot{N}_{ti}c^k$	$\gamma = 3$ (hand-set)	Only if $\gamma = 3$
Holographic count equation	$\mathbf{N_h = N_{ti}c^k}$	**None**	**Yes (derived)**
Constrained action	$S_e$ff with $\Lambda(t)$	None	Yes (variational)

Table 2: Four formulations of the SFT cosmological law. The holographic count equation and the constrained action are the native formulations.

7. Open Problems

Cosmic acceleration. The count equation with constant parameters produces $H^2 \propto \rho \propto (1+z)^3$, matching the V7.1 amplitude but not producing acceleration ($w < -1/3$). The transition from deceleration to acceleration requires z-dependent evolution of the count balance—likely through the screening mechanism or through the passive architectural channel identified in the bimodal model.

The second 2π from first principles. Section 4.2 identifies the second factor of 2π as the ratio between boundary thermodynamic normalization (Bekenstein-Hawking) and bulk field equation coupling (Einstein). A first-principles derivation showing that apparent-horizon thermodynamics *requires* this specific normalization ratio would complete the $4\pi^2$ story. Current status: identified but not yet derived.

Why the apparent horizon. In flat FLRW, the Hubble sphere coincides with the apparent horizon. This provides a rigorous thermodynamic justification for the boundary choice. Whether this identification generalizes to non-flat or inhomogeneous spacetimes is an open question.

Pressure and multi-component fluids. The derivation assumes pressureless dust. Radiation (massless, no rest-mass Compton frequency) and multi-component fluids require specifying how the count equation handles species with different equations of state.

Beyond minisuperspace. The constrained action (Eq. 8) is restricted to flat, homogeneous, isotropic spacetime. Extension to perturbations and fully inhomogeneous spacetimes is necessary for the framework to generate its own CMB and structure growth predictions natively, rather than through translation into CLASS.

8. Conclusion

Two results are established in this note:

(1) The holographic count equation—boundary cells equal bulk maintenance cycles over one Hubble time—provides a native SFT variational principle that produces $H^2 = 2G\rho/(3\pi)$ with zero free parameters. The constrained action $S_{eff} = \int dt\, \Lambda(t)[A_H/(4\ell_P^2) - E_H/(hH)] + S_{matter}$ is the mathematical foundation of the theory.

(2) The $4\pi^2$ ratio between SFT and standard Friedmann decomposes into two independent factors of 2π: one from the phase convention (h vs ħ) and one from the boundary-versus-bulk normalization (Bekenstein-Hawking vs Einstein coupling). Both are content-independent geometric ratios. Neither depends on the matter content of the universe.

Together, these results provide the mathematical skeleton for a future unification (V8) that will incorporate the screening mechanism, the bimodal entropy channels, and the z-dependent evolution needed to produce cosmic acceleration. The count equation is the foundation. The $4\pi^2$ decomposition is the geometric dictionary between SFT and GR. What remains is the architecture.